

24-hour Build Challenge

Build a harness — a framework that an AI agent lives inside. The harness defines what the agent gets for free: guardrails that constrain its behavior, checkpoints that evaluate its outputs, clean interfaces for passing material in and out, and alarms that fire when something goes wrong. What domain your agent operates in is your choice. What your harness provides, and how an agent works with it, is what you are being evaluated on.

Requirements

- **Must** All four pillars are implemented and demonstrably separate from the worker — guardrails, checkpoints, material handling, and alarms each exist as distinct, identifiable components in the code
- **Must** The harness governs an AI agent and the agent's behavior changes meaningfully based on guardrail or checkpoint feedback
- **Must** Guardrails are declared, not implicit. Checkpoints with explicit pass/fail criteria.
- **Must** Alarms produce structured output — named alarm types with context, severity, and a recommended action
- **Must** The harness runs on a real input from the engineer's own work at demo time
- **Must** An HARNESS.md file that covers the architecture and design of the harness
- **Should** Swappable agent interface — dropping in a different agent requires no changes to the harness
- **Should** Checkpoint results are persisted — you can replay a run from any checkpoint forward without re-running prior stages
- **Should** Contains human-in-the-loop escalation paths — the harness knows when to stop and ask rather than guess
- **Bonus** A second worker is swapped in during the demo to prove portability

Deliverables

Due Friday at 11:30 PM

- 1-page Harness Planning Document (Due Friday at 11:30 PM)

Due Saturday at 4:30 PM

- Project repo URL
- Deployed Harness URL
- Documentation on the harness's capabilities (saved as HARNESS.md in your code repo)
- 5-minute demo video on what you built and how it works

Agents focus on tasks. Harnesses focus on constraints. A well-designed harness makes constraint-handling invisible to the agent.